Exercise

2024年12月19日 20:14

线性回归

$$h_\theta(x) = \theta_0 + \theta_1 x$$

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta(x) - y(x)\right)^2 = \frac{1}{2m}\sum_{i=1}^{m}\left(\theta_0 + \theta_1 x^{(i)} - y(x^{(i)})\right)^2$$

$$\frac{\partial J(\theta)}{\partial \theta_0} = \frac{1}{m}\sum_{i=1}^{m}\left(\theta_0 + \theta_1 x^{(i)} - y(x)\right)$$

$$\frac{\partial J(\theta)}{\partial \theta_1} = \frac{1}{m}\sum_{i=1}^{m}\left(\theta_0 + \theta_1 x^{(i)} - y(x^{(i)})\right) x^{(i)}$$

$$\theta_0 = \theta_0 - \alpha\frac{\partial J(\theta)}{\partial \theta_0}$$

$$\theta_1 = \theta_1 - \alpha\frac{\partial J(\theta)}{\partial \theta_1}$$

多重特征值线性回归

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$$

令 $x_0 \equiv 1$，则 $h_\theta(x) = \theta_0 x_0 + \theta_1 x_1 + \cdots + \theta_n x_n = \vec{\theta}^T\vec{x}$

$$J(\vec{\theta}) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta(\vec{x}^{(i)}) - y^{(i)}\right)$$

$$\theta_j = \theta_j - \alpha\frac{\partial J(\vec{\theta})}{\partial \theta_j}$$

$$\frac{\partial J(\vec{\theta})}{\partial \theta_j} = \frac{1}{m}\sum_{i=1}^{m}\left[\underbrace{\left(h_\theta(\vec{x}^{(i)}) - y^{(i)}\right)}_{error^{(i)}} x_j^{(i)}\right], \quad x_0 \equiv 1$$

j 行 i 列

$$\frac{\partial J(\theta)}{\partial \theta_1} = \frac{1}{m}\sum_{i=1}^{m} error^{(i)} \cdot x_1^{(i)}$$

$$\frac{\partial J(\theta)}{\partial \theta_2} = \frac{1}{m}\sum_{i=1}^{m} error^{(i)} \cdot x_2^{(i)}$$

$$\Rightarrow \begin{bmatrix} \frac{\partial J(\theta)}{\partial \theta_1} \\ \frac{\partial J(\theta)}{\partial \theta_2} \\ \vdots \\ \frac{\partial J(\theta)}{\partial \theta_n} \end{bmatrix} = \frac{1}{m} X^T error$$

由于在处多特征，即不同特征间尺度差异可能比较大，因此，需要用到下面技巧：

均值归一化

$$u = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$x = \frac{x - u}{\max(x) - \min(x)}$$

x 变均值

归一化

正规方程

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}\left[h_\theta(x^{(i)}) - y^{(i)}\right]^2 = \frac{1}{2m}\left[X\theta - Y\right]^2$$

$$J'(\theta) = \frac{1}{m} X^T\left[X\theta - Y\right]$$

令 $J'(\theta) = 0$ 得 $X^T(X\theta - Y) = 0$，$\theta = (X^TX)^{-1}X^TY$ (若 X^TX 可逆)

又 X^TX 为正定阵，故 θ 为极小值点

逻辑回归

$y = 1$

$y = 0$

$h_\theta(x) = 0$ 时，$cost(h_\theta(x), y) = 0$

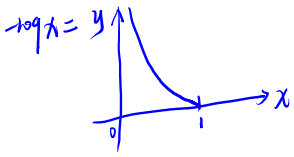
逻辑回归

损失函数设计：

$y=1$

$h_\theta(x)=0$ 时，$\text{cost}(h_\theta(x),y)=\infty$

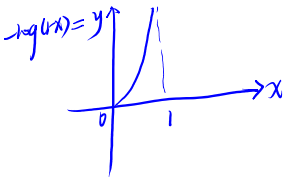
$h_\theta(x)=1$ 时，$\text{cost}(h_\theta(x),y)=0$



$y=0$

$h_\theta(x)=0$ 时，$\text{cost}(h_\theta(x),y)=0$

$h_\theta(x)=1$ 时，$\text{cost}(h_\theta(x),y)=\infty$



$$J(\theta)=\text{cost}(h_\theta(x),y)=-\frac{1}{m}\left[\sum_{i=1}^{m} y^{(i)}\log h_\theta(x^{(i)})+(1-y^{(i)})\log(1-h_\theta(x^{(i)}))\right]$$

$$\frac{\partial J(\theta)}{\partial \theta_j}=\frac{1}{m}\sum_{i=1}^{m}\left[(h_\theta(x^{(i)})-y^{(i)})\cdot x_j^{(i)}\right]$$

$$=\frac{1}{m}\left(\sum_{i=1}^{m} error^{(i)}\cdot x_j^{(i)}\right)$$

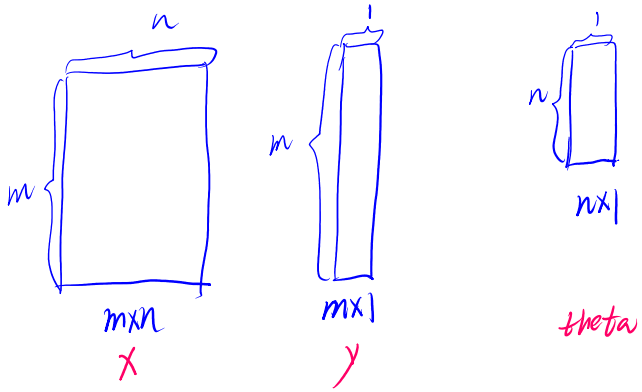
$$\begin{bmatrix}\frac{\partial J(\theta)}{\partial \theta_1}\\ \frac{\partial J(\theta)}{\partial \theta_2}\\ \vdots\\ \frac{\partial J(\theta)}{\partial \theta_n}\end{bmatrix}=\frac{1}{m}\sum_{i=1}^{m} error^{(i)}\cdot \begin{bmatrix}x_1^{(i)}\\ x_2^{(i)}\\ \\ x_n^{(i)}\end{bmatrix}$$

第1列　第2列　第n列

$$\begin{bmatrix}(x^1)^T\\ (x^2)^T\\ \\ (x^n)^T\end{bmatrix}$$

第1行　第2行　第n行

$$\rightarrow X^T\cdot error$$



正则化

逻辑回归

$$J(\theta)=\frac{1}{m}\sum_{i=1}^{m}\left[-y^{(i)}\cdot\log(h_\theta(x^{(i)}))-(1-y^{(i)})\cdot\log(1-h_\theta(x^{(i)}))\right]+\frac{\lambda}{2m}\sum_{j=1}^{n}\theta_j^2$$

$$\frac{\partial J(\theta)}{\partial \theta_j}=\frac{1}{m}\sum_{i=1}^{m}\left[h_\theta(x^{(i)})-y^{(i)}\right]+\frac{m}{\lambda}\theta_j$$